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Middle School STEM
Mary, Star of the Sea School
Innovations Lab (C-17)
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MSOS 2026-2027

STEM — Grade 7 Syllabus

Grade 7: Life Science • SY 2026-2027

Mary, Star of the Sea School Mission

Nurture, Unite, Inspire. We are committed to nurturing each child's curiosity, uniting our classroom in respect and collaboration, and inspiring every student to grow in knowledge, faith, and wonder for the world around them.

2026-2027 Essential Question

"How can I model kindness, fairness, and humility like Jesus?"

Course Description

7th Grade STEM is designed to develop scientifically literate, inquisitive students with a solid understanding of basic life science concepts and inquiry skills. It is focused on ways to unite our world view as citizens of the planet, nurture our understanding of how the biological systems are interrelated and inspire the next generation of scientists and engineers.

The year begins with a short review of the scientific method and the engineering design process as well as the principal tools and techniques used by scientists and engineers. Then we launch into a study of how we perceive the world. We will investigate how our senses interface with our brain, how the brain works, grows, and learns. Each student will conduct a dissection of an organ to gain first-hand knowledge of dissection skills. After Winter Break, our students will learn about the form and functions of cells, focusing on the role of bacteria and viruses in our world. This unit culminates in a formal debate about whether viruses are alive. Following Spring Break, the year continues with a study of human body systems, heredity, and genetics and ends with an examination of human reproduction and life in the womb.

Teacher

Dr. Dave Eachus

Innovations Lab (C-17)

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Classroom & Lab Materials

- Composition notebook (lab journal)
- Pencils, pens, colored pencils
- Lab safety goggles
- Computer / device for simulations
- Calculator

Scope & Sequence — Grade 7: Life Science

2026–2027, August–June. All unit dates are projected and may be subject to change.

Unit 1 — Why STEM? 3 Aug – 28 Aug

Key Concept: Systems · Related: Patterns, Form, Function · Global Context: Science and Technical Innovation

Inquiry statement: The Scientific Method and the Engineering Design Process allow us to investigate the form and function of patterns observed in natural systems or designs.

Factual: How do scientists and engineers use the Scientific Method and the Engineering Design Process?

Conceptual: How does experimental science and engineering expand our body of knowledge?

Debatable: Can the Scientific Method or the Engineering Design Process be used to answer any question or solve any problem?

Know: Scientific Method, Engineering Design Process, Human Water Cycle

Understand: How the Scientific Method is used to answer questions. The linkage between the natural water cycle and the human water cycle.

Do: Lab Safety, Scientific Method/Engineering Design Process, variables and hypothesis formulation, Lab report format, Water Quality Lab.

Summative Assessment Tasks:

- Hawaiian Water Cycle Poster (Crit. A/D)

Unit 2 — By the Dawn's Early Light 31 Aug – 18 Dec

Key Concept: Perspective · Related: Evidence and Patterns · Global Context: Identities and Relationships

Inquiry statement: A personal perspective is formed from learning based on evidence, facts, observations, and ideas.

Factual: How does your brain make sense of the world?

Conceptual: How does evidence shift perspective?

Debatable: How does your perspective relate to your values (perception of right or wrong)?

Know: Anatomy and physiology of the human brain and eye.

Understand: The uniqueness of each of our brains, responses by the brain differ between individuals, learning is an important brain function, and there are factors that affect learning, the ability of the brain to learn is not fixed.

Do: Dissection skills, designing, conducting, and communicating the results of a scientific investigation, collecting, interpreting, and analyzing data. Labs: Mass, Volume, Density, Learning/Memory Brain Investigation.

Summative Assessment Tasks:

- Technology and Perception Essay (Crit. A/D)
- Brain Memory/Learning Inquiry (Crit. B/C)

Unit 3 — An Invisible Army 7 Jan – 12 Mar

Key Concept: Relationships · Related: Function and Models · Global Context: Scientific and Technical Innovation

Inquiry statement: Examining the relationships between organisms allows us to model their functions to better understand how they interact.

Factual: What are the differences between plant, animal, and bacterial cells?

Conceptual: How do bacteria affect humans and the environment?

Debatable: Who is most likely to survive: humanity or bacteria?

Know: Cell organelles and functions, the differences among plant, animal, and bacterial cells, the differences between bacteria and viruses.

Understand: The process of osmosis, bacteria are both beneficial and harmful.

Do: Microscope skills, designing, conducting, and communicating the results of a scientific investigation, research skills. Labs: Respiration/Combustion Lab, Osmosis Investigation.

Summative Assessment Tasks:

- Cell Analogy Poster (Crit. A/D)
- Osmosis Lab (Crit. B/C)
- Virus Debate (Crit. A/D)

Unit 4 — Body Systems, Genetics & Reproduction 29 Mar – 3 Jun

Key Concept: Change · Related: Interaction and Transformation · Global Context: Orientation in Space and Time

Inquiry statement: Observing changes in interactions allows scientists to detect and understand the transformations in living organisms.

Factual: What is DNA and how does it work?

Conceptual: How does genetics relate to evolution?

Debatable: Does evolution discount the existence or workings of God?

Know: Human Body Systems, Mendel's Laws of Heredity, DNA/RNA transcription, Protein synthesis, evidence of evolution.

Understand: How the human body systems function to support each other. How genetics is the underpinning of the evolution of life on Earth.

Do: Body Systems, Blood Lab, Rice Babies, Punnett Squares, Sesame Street Genetics.

Summative Assessment Tasks:

- Rice Babies (Crit. A/D)
- Blood Lab (Crit. B/C)
- Sesame Street Genetics (Crit. A/D)

STEM Assessment Criteria

Each criterion is scored on an MYP-style scale of 0–8. Student activities are assessed against the criteria below; end-of-quarter letter grades reflect overall progress.

Criterion	Focus	Description	Scale
A	Knowing & Understanding	Recall and apply scientific knowledge; explain phenomena.	0–8
B	Inquiring & Designing	1st half of the Experimental / Design Cycle: question, plan, hypothesize, design.	0–8
C	Processing & Evaluating	2nd half of the Experimental / Design Cycle: collect, analyze, draw conclusions.	0–8
D	Reflecting on the Impacts of Science / Engineering	Consider ethical, environmental, and social implications of science and engineering decisions.	0–8

Learning Support & Communication

Parents with questions or concerns may directly contact the instructor.

Students are encouraged to use study hall or lunch recess to obtain additional learning support.

The Innovations Lab is open every Wednesday afternoon (except Mini-days) until 4 pm for any student needing help or to finish lab work.

Students and/or parents should email Dr. Eachus to make arrangements for additional learning opportunities.

Empowered to Succeed, Guided by Faith